This spreadsheet contains various test cases for the functions: TREND
Name of the main junit test class which uses this spreadsheet:
org.apache.poi.ss.formula.functions.TestTrendFunctionsFromSpreadsheet
(The content of cell 'Read Me'!A3 is confirmed by the test)

Every sheet besides this first one contains formula evaluation test cases in a standardis
On row 4 of each sheet, in columns B,C,D there are the column headings "Formula", "Expect
The test iterates down from row 5 onward until a the text "<end>" is found in column A.
Rows with "<skip>" in column A are ignored (useful for currently unsupported behaviour).
Otherwise column A can be used for commenting the group of rows below.
If the evaluated result of column B does not match the expected result in column C, the
junit test will report a failure. Test failures get annotated with the section and row
Besides the first 4 columns, row 5 onwards, the junit test does not inspect any other
cells in the sheet. The other cells can be used freely to set up test data.
Care should be taken to not only make the values in column C match those in column B, but
also to make sure column C contains only simple literal values. This can be achieved
easily by 'pasting special' from column B to C, selecting the 'as values' option.

ed Result" and "Comment"

Sheet Comment: Trend tests

| Formula | Expected Result |
|---|---|

Basic functionality

| | Formula | Expected Result |
|---|---|---|
| | 32.01313976 | 32.01313976 |
| | 3477.647388 | 3477.647388 |
| | -1455.227778 | -1455.227778 |
| | 1733.583694 | 1733.583694 |
| | 1554.468635 | 1554.468635 |
| | 13.376026 | 13.376026 |
| | 12 | 12 |
| | 5 | 5 |
| | 1.594595 | 1.594595 |
| | -2791.04684 | -2791.04684 |
| | 1733.583694 | 1733.583694 |
| | -3087.920731 | -3087.920731 |
| | 12 | 12 |
| | -3970.014038 | -3970.014038 |
| | 211.3844587 | 211.3844587 |
| | 217.033333 | 217.033333 |
| | 5 | 5 |
| | 13.54125356 | 13.54125356 |
| | 3509.703897 | 3509.703897 |
| | 20.924504 | 20.924504 |
| | 1733.335599 | 1733.335599 |

Not yet implemented

| | | |
|---|---|---|
| <skip> | 12 | 12 |
| <skip> | 12 | 12 |
| | 12 | 12 |
| | 5 | 5 |
| <skip> | -20 | -20 |
| <skip> | 57 | 57 |
| <skip> | 0 | 0 |
| <skip> | 5 | 5 |
| <skip> | 12 | 12 |
| <skip> | #REF! | #REF! |

Corner cases

| | | |
|---|---|---|
| | 3 | 3 |
| | 0.122881 | 0.122881 |
| | #REF! | #REF! |
| | 4031.422222 | 4031.422222 |
| | 32.01313976 | 32.01313976 |
| | | |
| | #VALUE! | #VALUE! |
| | #VALUE! | #VALUE! |
| | #VALUE! | #VALUE! |
| | #REF! | #REF! |

Array functions

| | | |
|---|---|---|
| | 23.119431 | 23.119431 |
| | 20.480586 | 20.480586 |
| | 22.379568 | 22.379568 |
| | 20.488807 | 20.488807 |
| | 22.379568 | 22.379568 |
| | 20.497028 | 20.497028 |
| | 23.119431 | 23.119431 |
| | 22.379568 | 22.379568 |
| | 22.379568 | 22.379568 |

〈end〉

| | | | |
|---|---|---|---|
| | | Database | |
| | | 12 | 24 |
| **Comment** | | 213 | 24 |
| | | 4 | 654 |
| | | 23423 | 234 |
| Forcing predicted line to pass through origin | | 423 | 567 |
| No new x values given; Note: rounded | | 423 | 57 |
| Only y and new x values given | | 532 | 324 |
| Multiple columns in area for x and y values; Note: rounded | | 2342 | 234 |
| Multiple columns again | | 323 | 234 |
| Single rows for x and y values; Note: rounded | | | |
| Reference same cells for both x and y values; Note: rounded | | | |
| Constant value for x and y + forcing line to pass through or | | 34 | 23 |
| Negative values; Note: rounded | | 324 | 123 |
| Multiple x variables for each y; Note: rounded | | | sa |
| Multiple columns for x and y values; Note: rounded | | 443 | 4 |
| Even more x variables for each y; Note: rounded | | | |
| Multiple x variables for each y (this time with rows) | | | |
| Another test with a given range for new_x's; Note: rounded | | | |
| Multiple columns for x's and new_x's + passing through origin | | | |
| Multiple columns for x's and new_x's; Note: rounded | | | 12 |
| Multiple same y values | | | 123 |
| Rows for x and y + new_x's different shape | | | 0 |
| Different shape for new_x's; Note: rounded | | | 3 |
| Another different shape for new_x's; Note:rounded | | | 13 |
| Multiple columns for x and y and different shape for new_x's | | | |
| | | | 5 |
| | | | 5 |
| | | | 5 |
| | | | |
| | | | -5 |
| | | | 6 |
| | | | -20 |
| | | | |
| | 12 | 24 | 5 |
| | 213 | 24 | 45 |
| | 4 | 654 | 1 |
| | 23423 | 234 | 45 |
| | 423 | 567 | 4 |
| | 423 | 57 | 23 |
| | 532 | 324 | 3 |
| | 2342 | 234 | 4 |
| | 323 | 234 | 5 |
| | 24 | | 2 |
| | 24 | | 32 |
| | 654 | | 34 |
| | 234 | | 34 |
| | 567 | | 32423 |
| | 57 | | 42 |
| | 324 | | 44 |

| | | | | |
|---|---|---|---|---|
| | 234 | | | 1 |
| | 234 | | | 2 |
| | 1733.58369 | 19.4099697 | #REF! | |
| | 1733.58369 | 19.4099697 | | |
| Only one given y value + forcing line to pass through origin; Not yet working in cu | | | | |
| Only one y value; Not yet working in current version of math library | | | | |
| All x values the same; Not working in math libra | 12 | 24 | | |
| All x's and y's the same; Not working in math lil | 12 | 24 | | |
| Negative values + only one value for x and y valu | 12 | 654 | | |
| Only one value for x, y, and new x values; Not ye | 12 | 234 | | |
| Zero values for x and y; Not yet working in curre | 12 | 567 | | |
| Constant value for x and y; Not yet working in cu | 12 | 57 | | |
| More x variables than samples | 12 | 324 | | |
| Passing new x value + only one y value is given; | 12 | 234 | | |
| | 12 | 234 | | |
| | 12 | 12 | | |
| | #REF! | 0.66224373 | | |
| | | | | 5 |
| | | | | 5 |
| Special matrices; Note: rounded | | | | 5 |
| More duplicate x values; Note: rounded; this one probably works because | | | | 5 |
| More rows for x than y values | | | | |
| Use named area instead of range; Note: this test fails due to minor rou | | | | 3 |
| Named area + forcing line to pass through origin | | | | 3 |
| | | | | 20 |
| Empty cells | | | | 4 |
| String instead of number | | | | |
| Only x values given | | | | 0 |
| Different shapes for x and y values | | | | |
| | | 1 | | 1 |
| | | 3 | | 2 |
| | | 4.5 | | 3 |
| | | 6 | | 4 |
| Array function with column | | 7.5 | | 5 |
| Array function with column | | 1 | | 2 |
| Array function with column | | 6 | | 4 |
| Array function with column | | 9 | | 6 |
| Array function with column | | 12 | | 8 |
| Array function with column | | 15 | | 10 |
| Array function with different shape for new_x's | | | | |
| Array function with different shape for new_x's | | | | 0.12288114 |
| Array function with different shape for new_x's | | | | |
| | | | | 3 |
| | | | | 3 |

3
3

5
5
5

| | | | | | | |
|---|---|---|---|---|---|---|
| | | 234 | | | | |
| 5 | 2 | 45 | | | | |
| 45 | 32 | 34 | #REF! | | | |
| 1 | 34 | 3509.7039 | 3514.76545 | 3509.7039 | 12 | 4 |
| 45 | 34 | 3509.7039 | 3514.76545 | 3509.7039 | 213 | 23423 |
| 4 | 32423 | 3509.7039 | 3514.76545 | 3509.7039 | 24 | 654 |
| 23 | 42 | 3509.7039 | 3514.76545 | 3509.7039 | 24 | 234 |
| 3 | 44 | 3509.7039 | 3514.76545 | 3509.7039 | 5 | 1 |
| 4 | 1 | 21.0247236 | | | 45 | 45 |
| 5 | 2 | 76.3333333 | 23.1194309 | 23.1194309 | | |
| | #REF! | 0.01834862 | 23.9332801 | | 12 | 4 |
| | 333.962359 | 333.962359 | 333.962359 | | 24 | 654 |
| | 301.791224 | 301.791224 | 301.791224 | | 5 | 1 |
| | 303.399781 | 333.962359 | | | 213 | 23423 |
| | 305.008338 | 301.791224 | | | 24 | 234 |
| | 333.962359 | | | | 45 | 45 |
| | 301.791224 | | | | 23.1194309 | 23.1194309 |
| | 303.399781 | | | | 20.4805864 | 22.3795679 |
| | 305.008338 | | | | 22.3795679 | 22.3795679 |
| | #N/A | | | | 20.4888071 | 20.4805864 |
| 0 | 333.962359 | 333.962359 | 333.962359 | 333.962359 | 22.3795679 | 20.4888071 |
| 123 | 301.791224 | 301.791224 | 301.791224 | 301.791224 | 20.4970278 | 20.4970278 |
| 45 | 303.399781 | 303.399781 | 303.399781 | 303.399781 | | |
| 5 | 305.008338 | 305.008338 | 305.008338 | 305.008338 | | |
| 7 | | | | | | |
| | 11.0013761 | 10.6481413 | 10.6873896 | 10.663093 | 3477.64739 | 12 |
| 5 | 11.0013761 | 10.6481413 | 10.6873896 | 10.663093 | | 12 |
| 5 | 11.0013761 | 10.6481413 | 10.6873896 | 10.663093 | | 12 |
| 5 | 11.0013761 | 10.6481413 | 10.6873896 | 10.663093 | | 12 |
| | | | | | | 12 |
| -7 | | | | | | 12 |
| -4 | | | | | | 12 |
| 0 | | | | | | 12 |
| 6 | | | | | | 12 |
| -4 | | | | | | 12 |
| | | | | | | 12 |
| | | | | | | 12 |
| 2 | 213 | 24 | 45 | 32 | | 12 |
| 32 | | | | | | 12 |
| 34 | 23423 | 234 | 45 | 34 | | 12 |
| 34 | | | | | | 12 |
| 32423 | | | | | | 12 |
| 42 | | | | | | 12 |
| 44 | | | | | | 12 |
| 1 | | | | | | 12 |
| 2 | | | | | | |
| | | | | | | |
| | 23.1194309 | | | | | |
| | 20.4805864 | | | | | |
| | 22.3795679 | | | | | |
| | 20.4888071 | | | | | |
| | 22.3795679 | | | | | |
| | 20.4970278 | | | | | |

23.1194309
22.3795679
22.3795679

ırrent version of math library

| | | | |
|---|---|---|---|
| 5 | 5 | 5 | |
| 5 | 5 | 4 | |
| 5 | 5 | 4 | |
| | 5 | | |
| | 3 | 2 | 4 |
| 3 | 6 | 4 | 8 |
| 3 | 9 | 6 | 12 |
| 3 | | | |
| 3 | | | |

| | | | |
|---|---|---|---|
| 2 | 3 | 10 | 12 |
| 67 | 6 | 5 | 4 |
| 7 | 6 | 4 | 123 |
| 6 | 7 | 7 | 32 |
| 5 | 4 | 2 | 23 |
| 2 | 3 | 4 | 12 |
| 67 | 6 | 5 | 4 |
| 7 | 6 | 4 | 123 |
| 6 | 7 | 7 | 32 |
| 5 | 4 | 2 | 23 |

| | |
|---|---|
| 1 | 0.1088312 |

| | |
|---|---|
| 3 | 3 |
| 3 | 3 |

3 3

5
5
5

| | | |
|---|---|---|
| 324 | | |
| 234 | | |
| 234 | | |
| 3 | | |
| 4 | | |
| 5 | | |
| | | |
| 324 | | |
| 3 | | |
| 234 | | |
| 4 | | |
| 234 | | |
| 5 | | |
| 23.1194309 | 20.4805864 | #N/A |
| 22.3795679 | 20.4888071 | #N/A |
| 22.3795679 | 20.4970278 | #N/A |
| #N/A | #N/A | #N/A |
| #N/A | #N/A | #N/A |
| #N/A | #N/A | #N/A |

| | |
|---|---|
| 3477.64739 | 5 |
| 3477.64739 | 5 |
| 2414.72103 | 5 |
| 3123.3386 | 5 |
| 2561.5061 | 5 |
| 3421.97029 | 5 |
| 2971.49198 | 5 |
| 3123.3386 | 5 |
| 3123.3386 | |
| #N/A | |
| #N/A | |
| #N/A | |
| #N/A | |
| #N/A | |
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| #N/A | |